

Lecture 11

VLSI Interconnect

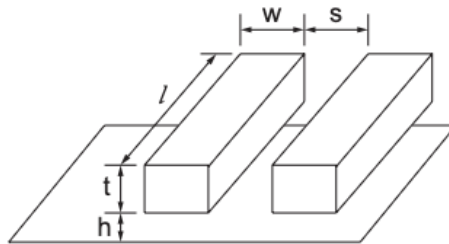
Prof Peter YK Cheung

Department of Electrical & Electronic Engineering

URL: www.ee.ic.ac.uk/pcheung/teaching/EE4-VLSI/
E-mail: p.cheung@imperial.ac.uk

Interconnect Wires

- ◆ **Wires** dominates VLSI chips.
- ◆ As feature sizes shrink → wires as important as transistors.
- ◆ Wires affect: 1) Speed; 2) Power; 3) Noise.
- ◆ Always use alternating layers at right angle to each other.
- ◆ Wire's pitch = $w + s$.
- ◆ Wire's aspect ratio $AR = t/w$.
- ◆ Modern process $AR \approx 2$.



W&H Ch 6

In modern VLSI processes, transistors switch much faster while wires have become narrower. As a result the **wire RC delay often exceeds gate delay**.

Moreover, large chip have lots of wires packed closely together. When one wire switches, signals can be capacitatively coupled to its neighbouring wire as **noise**.

Wires also account for a large portion of the **switching energy** of a chip.

On-chip **interconnect inductance** is now becoming a factor for systems with fast edge rates and closely packed busses. Therefore, circuit design is now as much about **engineering the wires** as the transistors that sit underneath.

The figure in the slide shows a pair of adjacent wires. The wires have width w , length l , thickness t , and spacing of s from their neighbours and have a dielectric of height h between them and the conducting layer below. The sum of width and spacing is called the **wire pitch**. The thickness to width ratio t/w is called the **aspect ratio**.

Metal Layer Stacks

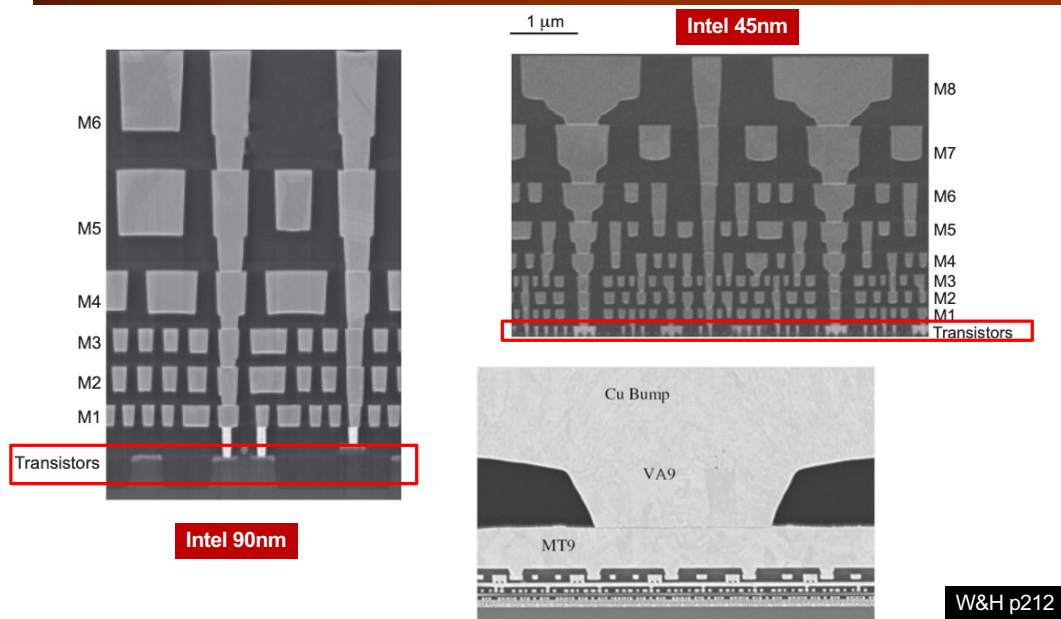
- ◆ A typical 0.6μm process has 3 metal layers:
 - ❖ M1 for within cell routing
 - ❖ M2 for vertical routing between cells
 - ❖ M3 for horizontal routine between cells.
- ◆ Modern processes use 6 to 10+ metal layers:
 - ❖ M1 for high density cells – thin and narrow
 - ❖ Middle layers (M2 up) – thicker and wider for routing between cells
 - ❖ Top few layers - for VDD, GND and clock signals.

Aluminum (Al) wires used in older processes. From around 180 nm node, **copper** (Cu) is used to reduce resistance.

Soon after, the introduction of copper wires, manufacturers began replacing the SiO₂ insulator between wires with a succession of materials with **lower dielectric** constants (**low-k**) to reduce capacitance.

A 65 nm process typically has 8–10 metal layers (our **TSMC65nmLP** process has 9 layers), and the layer count has been increasing at a rate of about one layer every process generation or two.

SEM images of metal stack



The Scanning Electron Microscope (SEM) images above show cross-sections of the metal stacks in the Intel 90 and 45 nm processes in mid 2000s.

The 90 nm process has six metal layers, while the 45 nm process has eight metal layers. Note how small are the transistor layer in comparison with the metal layers.

Metal1 is on the tightest pitch providing dense routing within cells.

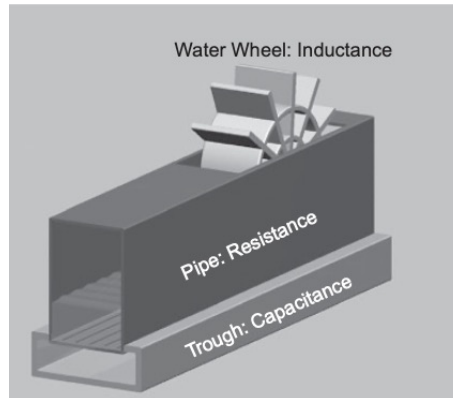
The upper levels are progressively thicker and on a greater pitch to offer lower resistance interconnections over progressively longer distances. The wires have a maximum aspect ratio of about 1.8. The two three layers are usually used for power, ground and clock routing to ensure low resistance.

The lower bottom image shows a full cross-section of Intel's 45nm process. This includes the very top layer with a Cu bump for connecting to the power or ground network in the package. and a VA9 via between MT9 and the bump. The lower levels of metal. The table below lists the metal layer dimensions for this process.

Layer	t (nm)	w (nm)	s (nm)	pitch (nm)
M9	7 μm	17.5 μm	13 μm	30.5 μm
M8	720	400	410	810
M7	504	280	280	560
M6	324	180	180	360
M5	252	140	140	280
M4	216	120	120	240
M3	144	80	80	160
M2	144	80	80	160
M1	144	80	80	160

Interconnect Modelling – waterpipe analogy

- ◆ Current in a wire is analogous to water in a pipe.
 - ❖ **Resistance**: narrow size impedes the flow of water.
 - ❖ **Capacitance**: trough under the leaky pipe must fill first.
 - ❖ **Inductance**: paddle wheel inertia opposes changes in flow rate (negligible except for long wires).



W&H p213

A pipe makes a good mechanical analogy for a wire, as shown above.

The **resistance** relates to the wire's cross-sectional area. A narrow pipe impedes the flow of current.

The **capacitance** relates to a trough underneath the leaky pipe that must fill up before water passes out the end of the pipe.

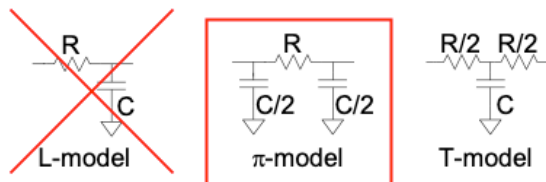
The **inductance** relates to a water wheel along the wire with inertia that opposes changes in the rate of flow.

Interconnect Modelling – Lumped model

- Wires are distributed circuits approximated by a **lumped element model**.



- 3-segment of π -model is accurate to 3% in simulation.
- L-model requires 100+ segments to achieve same accuracy.
- Use single 3-segment π -model for Elmore delay calculation.



W&H p213

A wire is a distributed circuit with a resistance and capacitance per unit length. Its behaviour can be approximated with a number of lumped elements. The upper circuit shows how a distributed RC circuit is equivalent to N distributed RC segments, of proportionally smaller resistance and capacitance.

Such distributed RC model can be approximated by lumped element models.

The **L-model** is a poor choice because a large number of segments are required for accurate results.

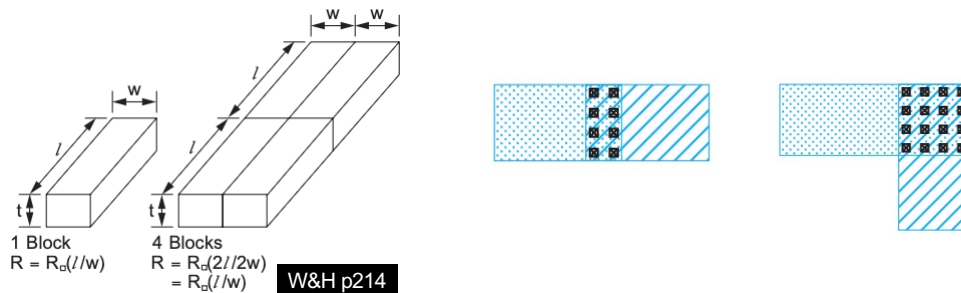
The **π -model** is most popular because only three segments are sufficient to give results accurate to 3% [Sakurai83].

The **T-model** is comparable to the π -model, but produces a circuit with one more node that is slower to solve by hand or with a circuit simulator. Therefore, it is common practice to model long wires with a 3 to 5 segment model for simulation.

If inductance is considered, it is placed in series with each resistor. Inductances are normally ignore except for very long interconnect wires.

Interconnect Resistance

- ◆ ρ = resistivity ($\Omega \cdot \text{m}$).
- ◆ Resistance $R = \frac{\rho}{t} \times \frac{l}{w}$.
- ◆ Sheet Resistance $R_{sq} = \frac{\rho}{t}$, dimension is ohms per square.
- ◆ Count number of squares: $R = R_{sq} \times \text{no. of squares}$
- ◆ Two wires below have same resistance.
- ◆ Polysilicon and diffusion sheet resistance: much higher at $10 \Omega/\text{sq}$ to $100\text{s } \Omega/\text{sq}$.
- ◆ Contact and vias typically has $2 - 20 \Omega$.



In calculating the resistance of wires, we use sheet resistance as defined above.

To calculate the wire resistance, we only need to count the number of squares and multiply it by the sheet resistance.

Above also shows the resistivity of different types of metals at 22°C .

The resistivity of polysilicon, diffusion, and wells depends on the doping levels. Polysilicon and diffusion typically have sheet resistances under $10 \Omega/\text{square}$ when silicided and up to several hundred Ω/square when unsilicided. Wells have lower doping and thus even higher sheet resistance. Therefore avoid using wells for interconnection unless you want large resistors (as in analogue circuits).

Contacts and vias also have a type resistance of $2-20\Omega$. Multiple contacts should be used to form low-resistance connections as shown in the slide.

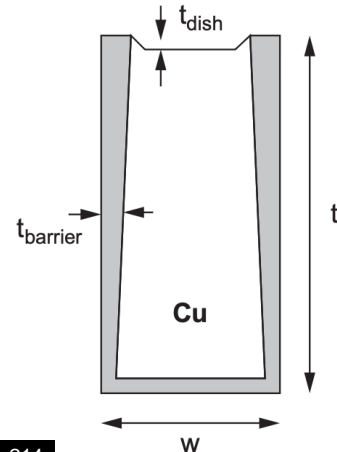
When current turns at a right angle, use a square array of contacts. If the flow of current is in the same direction, fewer contacts can be used.

Copper wires

- ◆ Copper wires diffusion barrier has high resistance
- ◆ Copper is also prone to *dishing* during polishing
- ◆ Effective resistance is higher

$$R = \frac{\rho}{(t - t_{\text{dish}} - t_{\text{barrier}})} \frac{l}{(w - 2t_{\text{barrier}})}$$

Metal	Resistivity ($\mu\Omega \cdot \text{cm}$)
Silver (Ag)	1.6
Copper (Cu)	1.7
Gold (Au)	2.2
Aluminum (Al)	2.8
Tungsten (W)	5.3
Molybdenum (Mo)	5.3
Titanium (Ti)	43.0



The above table shows copper (Cu) interconnect has very low resistivity, and is much cheaper than its competitor, Silver (Ag). Therefore copper interconnects were introduced since around 180nm process node.

However, copper has very high atomic mobility and it easily migrates into surrounding materials such as silicon dioxide. Such diffusion changes transistor characteristics, increase leakages, accelerate dielectric break down and generally accelerate degradation of the devices surrounding them.

The solution to the copper migration problem is to surround the copper wire with a barrier usually made of tantalum (Ta) or tantalum nitride (Ta₃N₂). Such barrier reduces the wire cross-section area and increase the resistance of the copper wire.

Furthermore, since copper is much softer than surrounding dielectric.

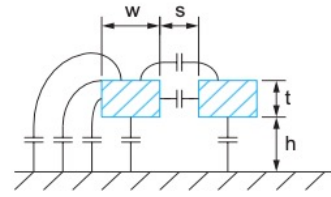
Another cause of increase resistance is called copper dishing. After a metal layer is created, it is subjected to a flattening process known as Chemical Mechanical Polishing (CMP). CMP ensures that successive layers of metal are flat, this allowing many layers of metal to be fabricated. Because copper is softer, polish create concave hollow on the copper surface as shown in the diagram. Even a 10 nm barrier is quite significant when the wire width is only tens of nanometers.

If the average barrier thickness is t_{barrier} and the height is reduced by t_{dish} , the resistance is reduce to:

$$R = \frac{\rho}{(t - t_{\text{dish}} - t_{\text{barrier}})} \frac{l}{(w - 2t_{\text{barrier}})}$$

Wire capacitances

- ◆ Parallel plate equation: $C = \frac{\epsilon_{ox}}{h} wl$
- ◆ Wires are not parallel plates
 - Increasing area (W, t) increases capacitance
 - Increasing distance (s, h) decreases capacitance
- ◆ Dielectric constant
 - $\epsilon_{ox} = k\epsilon_0$
 - $\epsilon_0 = 8.85 \times 10^{-14}$ F/cm
 - $k = 3.9$ for SiO_2
- ◆ Processes are starting to use low-k dielectrics
 - $k \approx 3$ (or less)



W&H p215

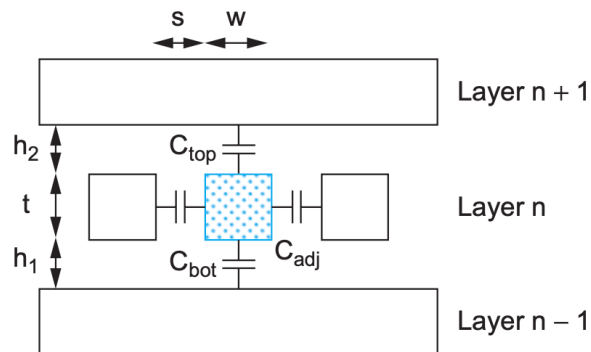
The wire capacitance has two major components: the parallel plate capacitance of the bottom of the wire to ground and the fringing capacitance arising from fringing fields along the edge of a conductor with finite thickness.

In addition, a wire adjacent to a second wire on the same layer can exhibit capacitance to that neighbour as shown in the diagram above.

Low-k dielectrics are used since 2003 with 130nm and 90nm nodes. With lower k value, the capacitances are reduced.

Wire capacitances

- ◆ Wire has capacitance per unit length
 - To neighbours
 - To layers above and below
- ◆ $C_{\text{total}} = C_{\text{top}} + C_{\text{bot}} + 2C_{\text{adj}}$

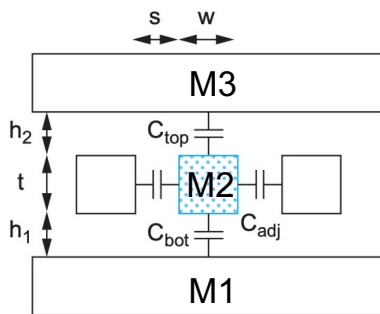


W&H p216

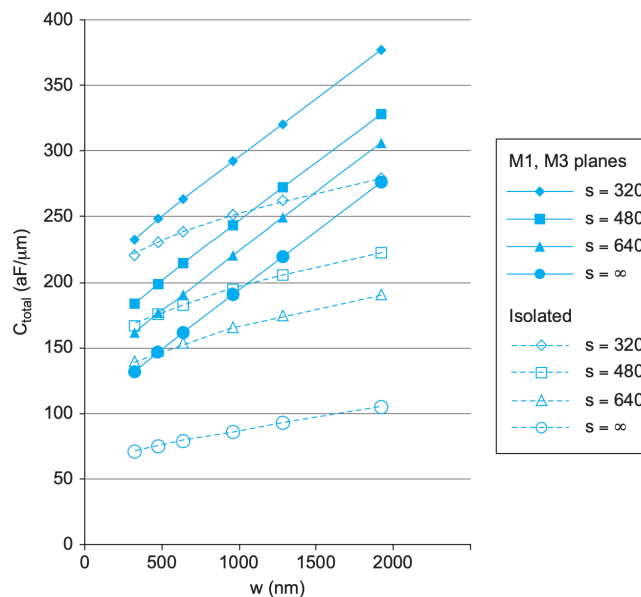
A cross-section of the model used for capacitance upper bound calculations is shown above. The total capacitance of the conductor of interest is the sum of its capacitance to the layer above, the layer below, and the two adjacent conductors.

Metal 2 capacitance

- ◆ Typical dense wires have $\sim 0.2 \text{ fF}/\mu\text{m}$
- ◆ Compare to 1-2 fF/ μm for gate capacitance



W&H p217



Wire capacitances of a given layer are therefore influenced by the metal interconnect layers above and below it.

In practice, the capacitances are computed by generating a lookup table of data with a field solver CAD tools. The table may contain data for different widths and spacings for each metal layer, assuming the layers above and below are occupied or unoccupied.

The graph in the slide above shows representative data for a metal 2 wire in a 180 nm process with wire and oxide thicknesses of $0.7 \mu\text{m}$. It plots the capacitance of M2 wire with different width w and separation s .

The solid lines show effect of M1 and M3 sandwich increasing the capacitance as compared to the hollow lines where the M2 layer is isolated.

Diffusion and Polysilicon Wires

- ◆ Diffusion capacitance is very high (1-2 fF/ μm)
 - Comparable to gate capacitance
 - Diffusion also has high resistance
 - Avoid using diffusion *runners* for wires!
- ◆ Polysilicon has lower C but high R
 - Use for transistor gates
 - Occasionally for very short wires between gates

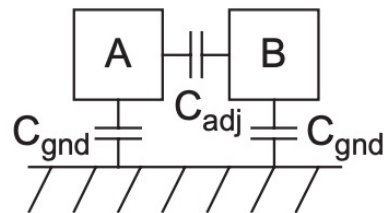
W&H p217

Never use diffusion as interconnect wires.

Keep polysilicon interconnect as short as possible.

Wire Crosstalk

- ◆ A capacitor does not like to change its voltage instantaneously.
- ◆ A wire has high capacitance to its neighbour.
 - When the neighbour switches from 1 $\rightarrow$ 0 or 0 $\rightarrow$ 1, the wire tends to switch too.
 - Called capacitive *coupling* or *crosstalk*.
- ◆ Crosstalk causes
 - Noise on non-switching wires
 - Increased delay on switching wires



W&H p222

Wires have capacitance to their adjacent neighbours as well as to ground. When wire A switches, it tends to bring its neighbor B along with it on account of capacitive coupling, also called crosstalk.

If B is supposed to switch simultaneously, this may increase or decrease the switching delay. If B is not supposed to switch, crosstalk causes noise on B.

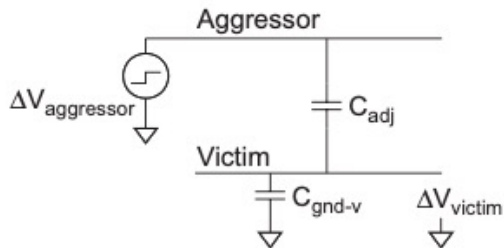
The load capacitance is included in the total capacitance calculation.

For short wires and large loads, the load capacitance dominates and crosstalk is unimportant.

Conversely, crosstalk is very important for long wires.

Crosstalk Noise – floating victim

- ◆ Crosstalk causes noise on non-switching wires
- ◆ If victim is floating:
 - model as capacitive voltage divider



$$\Delta V_{\text{victim}} = \frac{C_{\text{adj}}}{C_{\text{gnd-v}} + C_{\text{adj}}} \Delta V_{\text{aggressor}}$$

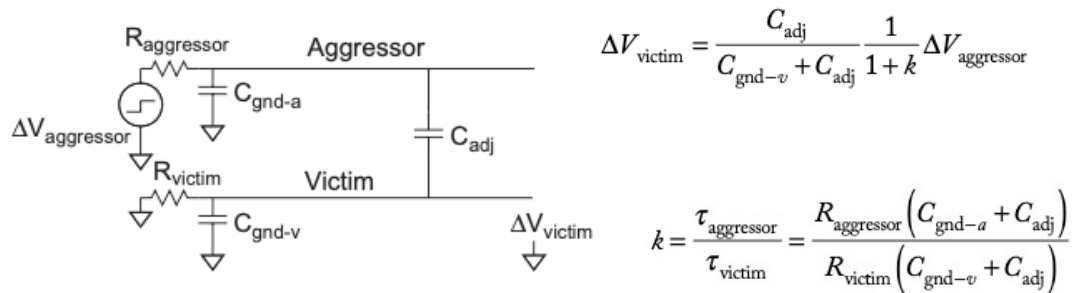
W&H p223

Suppose wire A switches while B is supposed to remain constant. This introduces noise as B partially switches. We call A the aggressor or perpetrator and B the victim.

If the victim is floating, we can model the circuit as a capacitive voltage divider to compute the victim noise as shown in the slide above.

Crosstalk Noise – driven victim

- ◆ Usually victim is driven by a gate that fights noise
 - Noise depends on relative resistances
 - Having driven victim increase k and reduces crosstalk noise
 - Avoid floating wires to minimize crosstalk



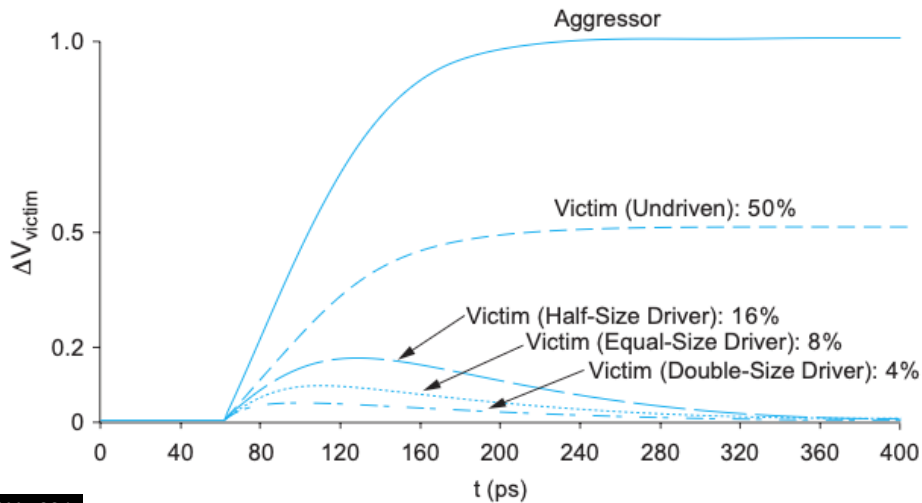
W&H p223

If the victim is actively driven, the driver will supply current to oppose and reduce the victim noise. We model the drivers as resistors. The peak noise becomes dependent on the time constant ratio k of the aggressor to the victim as shown in the equation in the slide above.

Therefore, try to avoid floating wires!

Noise Coupling Waveforms

- ◆ Simulated coupling for $C_{adj} = C_{victim}$



W&H p224

The graph above shows simulations of coupling when the aggressor is driven with a unit inverter; the victim is undriven or driven with an inverter of half, equal, or twice the size of the aggressor.

When the victim is floating, the noise remains indefinitely in time and is large.

When the victim is driven, the driver restores the victim. Larger drivers oppose the coupling sooner and result in noise that is a smaller percentage of the supply voltage.

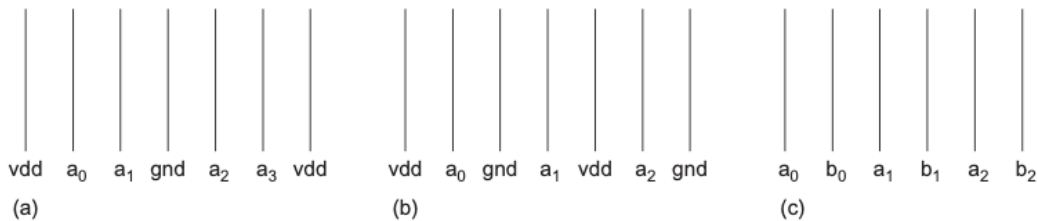
So, what if we have noise?

- ◆ If the noise is less than the noise margin, nothing happens.
- ◆ Static CMOS logic will eventually settle to correct output even if disturbed by large noise spikes.
 - Glitches cause extra delay.
 - Glitches cause extra power from false transitions.
- ◆ Dynamic logic never recovers from glitches.
- ◆ Memories and other sensitive circuits also can produce the wrong answer

If noise is below the noise margin of static logic gates, the circuit would still function but will incur higher power dissipation due to glitches and may increase the delay. However, noise can be catastrophic for dynamic logic and memory circuits because noise glitches can corrupt memory or falsely trigger dynamic pre-charged logic gates.

Crosstalk Control via Shielding

- ◆ Increase spacing to adjacent wires
- ◆ Shield wires
- ◆ Ensure neighbours switch at different times
- ◆ Crosstalk cancellation



W&H p233

The easiest approach to fix a minor crosstalk problem is to increase the spacing.

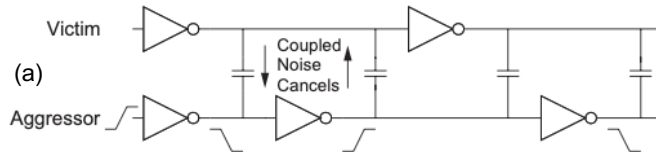
If the crosstalk is severe, it is more efficient to shield critical signals with power or ground wires on one or both sides to eliminate coupling. For example, clock wires are usually shielded so that switching neighbours do not affect the delay of the clock wire and introduce clock jitter. This is shown in (a) and (b) above.

Sensitive analogue wires passing near digital signals must always be shielded.

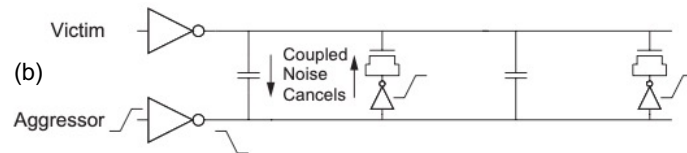
An alternative to shielding is to interleave busses that are guaranteed to switch at different times as shown in (c). For example, if bus A switches on the rising edge of the clock and bus B switches on the falling edge of the clock, by interleaving the bits of the two busses you can guarantee that both neighbours are constant during a switching event. This avoids the delay impact of coupling; however, you must still ensure that coupling noise does not exceed noise budgets.

Crosstalk Control via Cancellation

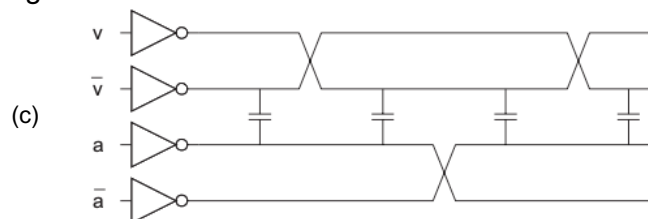
- ◆ Staggered repeaters:



- ◆ Charge compensation:



- ◆ Twisted differential signalling:



W&H p234

An alternative approach is to arrange the wires to cancel the effects of crosstalk. Each technique seeks to cause equal amounts of positive and negative crosstalk on the victim, effectively producing zero net crosstalk. These are:

- (a) Staggered repeaters - two wires with staggered repeaters. Each segment of the victim sees half of a rising aggressor segment and half of a falling aggressor segment.
- (b) Charge compensation - an inverter and transistor are added between the aggressor and victim. The transistor is connected to behave as a capacitor. When the aggressor rises and couples the victim upward, the inverter falls and couples the victim downward. By choosing an appropriately sized compensation transistor, most of the noise can be cancelled at the expense of the extra circuitry.
- (c) Twisted differential signalling - each signal is routed differentially. The signals are swapped or twisted such that the victim and its complement each see equal coupling from the aggressor and its complement. This approach is expensive in wiring resources, but it effectively eliminates crosstalk. It is widely used in memory designs that are naturally differential.